

M1 Garand Rifle

PAGES FROM THE ORIGINAL MANUAL

FM 23-5, Basic Field Manual – U.S. Rifle, Caliber .30, M1 (War Department, 1940)

WHAT'S INSIDE

Chapter 1, Mechanical Training: description, taking it apart and putting it back together, cleaning, how it works, operation, stoppages, spare parts and ammunition.

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WAR DEPARTMENT

BASIC FIELD MANUAL



U. S. RIFLE, CALIBER
.30, M1

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BASIC FIELD MANUAL

U. S. RIFLE, CALIBER .30, M1

(The matter contained herein supersedes chapter 1A, part one, Basic Field Manual, volume III, January 3, 1938.)

CHAPTER 1

MECHANICAL TRAINING

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SECTION I

GENERAL

■ 1. OBJECT.—This chapter is designed to give the soldier training that will insure his ability to maintain the rifle and keep it in operation.

■ 2. DESCRIPTION OF RIFLE.—The U. S. rifle, caliber .30, M1, is a self-loading shoulder weapon. (See fig. 1.) It is gas operated, clip fed, and air cooled. It weighs approximately 9 pounds and the bayonet an additional pound. The ammunition is loaded in clips of eight rounds. Bandoleers of ammunition for this rifle have six pockets with a total of 40 rounds and weigh $3\frac{1}{4}$ pounds each.

■ 3. FIREPOWER.—The principal characteristic of the weapon is its mechanical operation which enables the individual rifleman or group of riflemen to deliver a large volume of accurate fire upon any designated point or area within range.

SECTION II

DISASSEMBLY AND ASSEMBLY

■ 4. WHEN TAKEN UP.—This training will be taken up as soon as practicable after the soldier receives his rifle. In any case it will be completed before any firing is done with the rifle by the individual. Instruction in the care and cleaning of the rifle will also be covered.

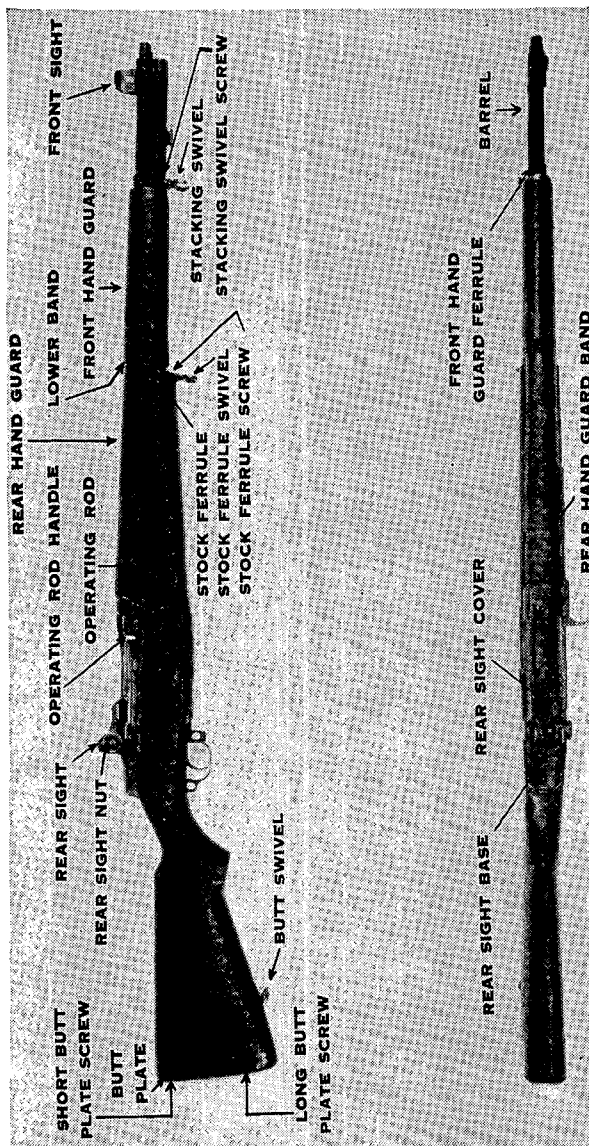


FIGURE 1.—U. S. rifle, caliber .30, M1.

■ 5. ORGANIZATION.—In the company* or platoon, all enlisted men are combined in one or more groups under their officers or selected noncommissioned officers as instructors. Other noncommissioned officers supervise the work as directed. Corporals supervise the work of their squads.

■ 6. CARE TO BE EXERCISED.—*a.* The rifle can be readily disassembled and assembled without applying force. Such practice is prohibited.

b. The piece will not be disassembled or assembled against time as this serves no useful purpose and results in burring and damaging the parts. Instruction, blindfolded, may be given to men who have passed their tests in mechanical training. In all work in disassembling the rifle the men will be taught to lay the parts out on a smooth, clean surface in the proper sequence for assembling.

■ 7. NOMENCLATURE.—The names of the parts to which reference is made in mechanical training are readily learned as this training progresses. Instructors will therefore take care to name the parts clearly and correctly in their work. A sufficient knowledge of the nomenclature of the rifle is gained by the soldier during the instruction in mechanical training.

■ 8. DISASSEMBLING.—*a. General.*—(1) Authorized disassembly by the soldier is limited to that required for proper care and maintenance of the rifle. Further disassembly will generally be done under the supervision of an officer or ordnance personnel. The individual soldier usually will be prohibited from—

- (a) Disassembling the stock group.
- (b) Disassembling the follower assembly.
- (c) Disassembling the rear sight.
- (d) Removing the clip latch.
- (e) Disassembling the trigger housing group.
- (f) Removing the gas cylinder lock.

(2) The front sight will not be dismounted at other than properly equipped Ordnance establishments.

*The term "company" as used in this manual includes troop, battery, or similar organization.

(3) The gas cylinder will not be disassembled from the rifle except by Ordnance personnel.

b. Sequence.—(1) The disassembly of the rifle authorized to be performed by the soldier without supervision is performed in the following sequence:

(a) *Dismounting three main groups.*

1. Remove trigger housing group.
2. Separate stock group from barrel and receiver group.

(b) *Disassembling barrel and receiver group.*

1. Remove follower rod with compensating spring attached.
2. Remove compensating spring from follower rod.
3. Remove operating rod spring.
4. Remove follower arm pin.
5. Remove bullet guide, follower arm, and operating rod catch assembly.
6. Remove follower with slide attached.
7. Remove operating rod.
8. Remove bolt assembly.

(c) *Disassembling bolt assembly.*

1. Remove extractor.
2. Remove extractor spring and plunger.
3. Remove ejector with ejector spring attached.
4. Remove firing pin.

(2) Disassembly of the following parts, generally prohibited for the individual soldier, is performed in the following sequence:

(a) *Removing clip latch.*

1. Remove clip latch pin.
2. Remove clip latch with clip latch spring attached.

(b) *Disassembling trigger housing group.*

1. Close and latch trigger guard.
2. Pull trigger.
3. Remove trigger pin.
4. Remove trigger assembly.
5. Remove hammer spring housing, hammer spring, and hammer spring plunger.
6. Remove hammer pin.
7. Remove hammer.

8. Remove safety.
 9. Remove trigger guard.
 10. Remove clip ejector.
- (c) *Removing gas cylinder lock.*
1. Remove gas cylinder lock screw.
 2. Remove gas cylinder lock.

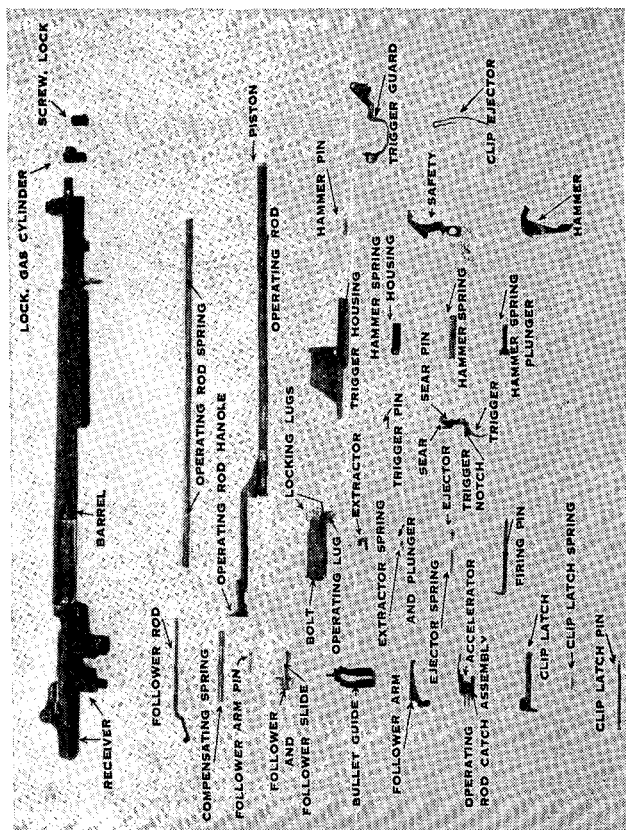


FIGURE 2.—Parts of U. S. rifle, caliber .30, M1.

c. Method.—The following detailed explanation of the method of disassembling the rifle is furnished as an aid to instructors:

(1) *Three main groups.*—Grasp rifle with left hand so that base of trigger housing is included in grip of fingers. (See fig. 3.) Place butt of rifle against left thigh. Grasp rear portion of trigger guard with thumb and forefinger of right hand and exert sufficient downward pressure to unlatch trigger guard from trigger housing. Then swing



FIGURE 3.

trigger guard away from trigger housing to extreme opened position as shown in figure 3. Pull out trigger housing group. Place this group on a smooth, clean surface. Grasp rifle over rear sight with left hand, muzzle down and barrel to left. (See fig. 4.) Strike and grasp small of stock with right hand so as to separate stock group from barrel and receiver group as shown in figure 4.

(2) *Barrel and receiver group.*—(a) Place group on a smooth surface, barrel down and muzzle to the left. Grasp

follower rod with left thumb and forefinger at knurled portion and disengage it from follower arm by pressure toward muzzle. Withdraw follower rod (with compensating spring attached) to the right. (See fig. 6.) If necessary, remove compensating spring from follower rod by grasping compensating spring with left hand and twisting follower rod toward body with right hand, exerting a slight pull to the right. Withdraw operating rod spring.



FIGURE 4.

(b) Drift follower arm pin from its seat by starting it with the point of a bullet (or with drift of combination tool) held in right hand and applied on side of receiver farthest from body. Pull out pin from near side with left hand. Grasp bullet guide, follower arm, and operating rod catch assembly and pull to the left until these parts are disengaged. Lift out and separate these three parts. Do not remove accelerator from operating rod catch assembly as accelerator pin is riveted in its seat. Lift out follower with

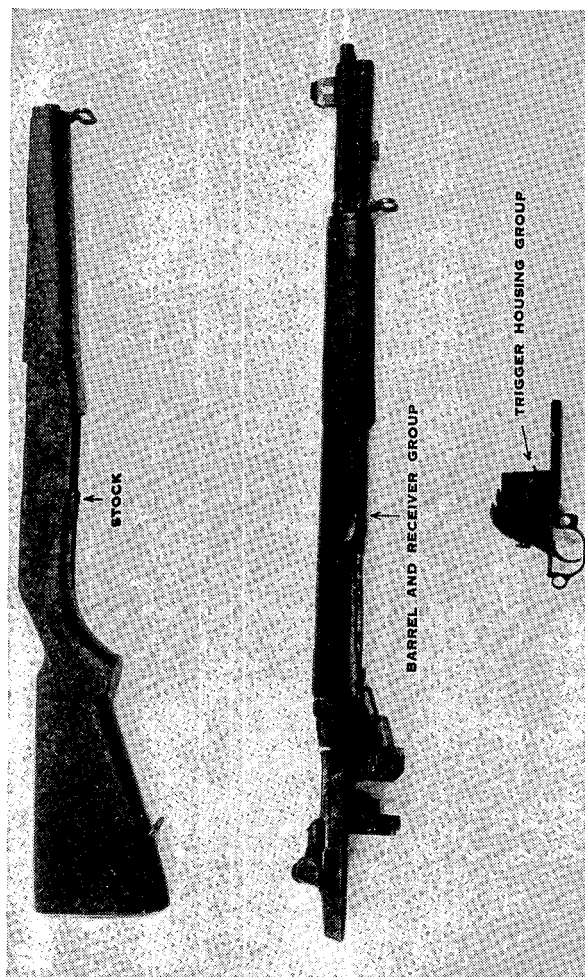


FIGURE 5.—Three main groups, U. S. rifle, caliber .30, M1.

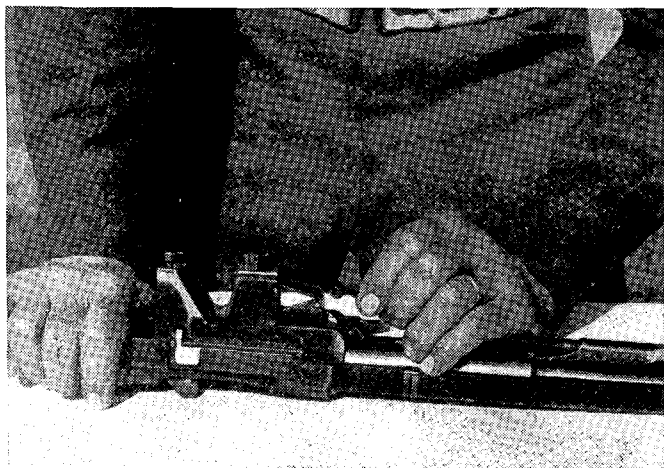


FIGURE 6.

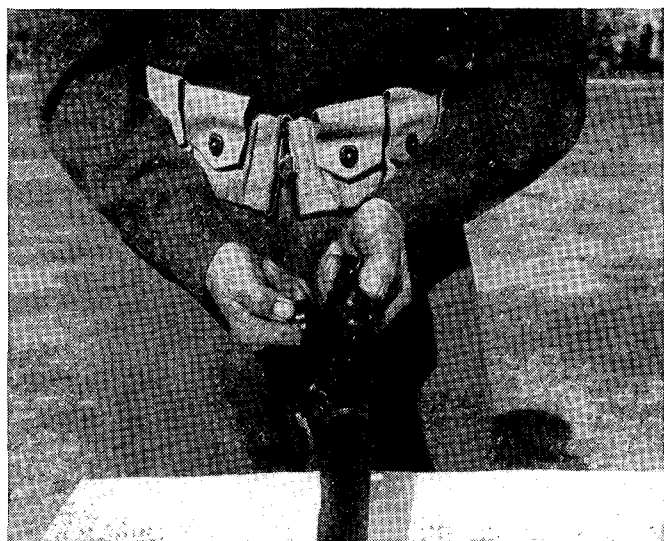


FIGURE 7.

follower slide attached. Do not separate follower from follower slide.

(c) Grasp barrel and receiver assembly with left hand and operating rod handle with right hand as shown in figure 7. Move operating rod slowly to the rear, pulling operating rod handle upward and away from receiver. This will disengage operating rod from bolt when lug on operating rod slides up into dismount notch of operating rod guide groove in



FIGURE 8.

receiver. When operating rod is thus disengaged remove it with a downward and rearward movement.

NOTE.—The operating rod has been intentionally bent. Do not attempt to straighten it.

(d) Grasp bolt by operating lug, slide it from rear to front, and lift it up and out to the right front with a slight rotary motion. (See fig. 8.)

(3) *Bolt assembly.*—Grasp bolt in left hand, holding firing pin in place with little finger, extractor to right, front end up,

thumb on front end of bolt to prevent ejector from flying out. (See fig. 9.) Insert point of a bullet or screwdriver blade of combination tool between lower edge of extractor and cartridge seat flange on bolt and pry extractor out until ejector snaps out against thumb. Extreme care must be exercised that ejector is held by thumb during this operation as ejector spring is compressed sufficiently to inflict injury if released. Remove extractor. Remove extractor spring and plunger. (Separate only when necessary.) Remove ejector with



FIGURE 9.

ejector spring attached. (Separate only when necessary.) Remove firing pin from rear of bolt.

(4) *Rear sight.*—(a) Turn barrel and receiver up. Lower aperture as far as it will go, noting reading on elevating knob and record it for use in assembling. (This reading should be approximately 100 yards.)

(b) Using combination tool, unscrew rear sight nut from right side of rear sight. Unscrew windage knob, taking care that rear sight nut lock assembly which is inside windage knob does not become lost. Remove nut lock and nut lock spring from windage knob.

(c) With a screwdriver remove rear sight elevating knob screw from left side of rear sight. Remove rear sight elevating knob. Pull out rear sight elevating pinion from left side of receiver. Grasp aperture and pull upward about $\frac{1}{2}$ inch, place thumb under top of aperture and push forward to remove rear sight cover, rear sight base, and aperture. Separate.

(5) *Clip latch.*—Place receiver on its right side with barrel to the left. Press clip latch with left thumb to relieve tension of clip latch spring, then drift out and remove clip latch pin, starting it toward muzzle with point of a bullet or drift of combination tool. Lift out clip latch with clip latch spring attached.

(6) *Trigger housing group.*—(a) The trigger housing group being removed, close and latch trigger guard. Release hammer to the fired position. Hold group in right hand, with right thumb on sear, forefinger pulling back on trigger, base of trigger housing braced against a firm support, and press on sear with right thumb to relieve tension on trigger pin. Start trigger pin from its seat with a drift held in left hand, then remove it. Release pressure of right thumb and forefinger gradually, permitting hammer spring to extend to its full length, at the same time steadying hammer spring housing with fingers of left hand. (See fig. 10.) Remove trigger assembly (do not remove sear pin or sear). Remove hammer spring housing, hammer spring, and hammer spring plunger, and separate these parts. Push out hammer pin from left side and remove hammer. Remove safety by pressing its top away from left side of trigger housing until stud snaps out of its seat, and lift it from its slot in trigger housing.

(b) Hold trigger housing in left hand, base of trigger housing down and away from body. Swing trigger guard to open position with right hand. Slide trigger guard toward body until hammer stop is opposite center of safety slot. Rotate trigger guard to right and upward with right hand until hammer stop clears edge of trigger housing. Remove trigger guard.

(c) Hold trigger housing in left hand, right side down, rear end resting on a solid surface. Insert point of a bullet

or combination tool in the dismounting hole in left side of housing and push out clip ejector.

(7) *Gas cylinder.*—(a) There are now in existence two types of gas cylinder and front sight assemblies for the M1 rifle. In the newer or spline type the barrel protrudes beyond the gas cylinder, and the front sight screw enters from the front and is sealed to prevent tampering. In the older or screw-on type the barrel does not protrude and the front sight screw enters from the side.



FIGURE 10.

(b) In order to prevent undue wear, insure proper maintenance of gas port adjustment, and avoid improper assembly, the gas cylinder assembly should not be removed except when necessary to replace the front hand guard assembly. Before such removal be sure the operating rod has been removed. To remove the gas cylinder, proceed as follows:

1. *Spline type.*—Unscrew gas cylinder lock screw using combination tool. Unscrew gas cylinder lock. Tap gas cylinder lightly toward muzzle to remove

it from barrel. The front sight will not be dismounted from the gas cylinder assembly. Do not attempt to adjust the front sight.

2. *Screw-on type.*—Remove front sight screw and lift out front sight. Unscrew gas cylinder assembly from barrel.

d. Disassembly of certain parts prohibited.—The repair or replacement of the following parts pertains to the Ordnance Department and their disassembly will not be effected within Infantry organizations: the trigger and sear assembly; the operating rod catch and accelerator assembly; the front sight; the gas cylinder.

■ 9. ASSEMBLING.—*a.* The rifle and its component groups are assembled in the reverse order of their disassembly as given in paragraph 8 b. The following detailed explanation of the method of assembling the rifle is furnished as an aid to instructors:

(1) *Gas cylinder lock.*—Replace gas cylinder lock and gas cylinder lock screw.

(2) *Trigger housing group.*—(a) Place clip ejector in position in trigger housing with short arm up, tip of long arm in its slot in vertical front face of housing, and loop against its stud on left side of trigger housing. Hold trigger housing with its left side down, rear end to the right. With right thumb hold loop of clip ejector against its stud; with forefinger of left hand hold long arm of clip ejector up in its slot in front face of trigger housing. With left thumb pry toward body on center of long arm of clip ejector, thereby seating it.

(b) Hold trigger housing in left hand, top up, forward end to front. Hold trigger guard in right hand, winged section pointing to the left. Place winged sections astride bottom of trigger housing, hammer stop over safety slot. Rotate trigger guard downward, then slide it forward into position.

(c) Replace safety so that its thumb piece passes through slot in both trigger guard and bottom of housing and its stud is snapped into its seat in side plate of trigger housing. Push thumb piece forward to ready position. Insert hammer loosely in cocked position. Aline pin hole in hammer with pin holes in trigger housing and trigger guard. Insert ham-

mer pin from the right. Swing hammer to the fired position. Place trigger housing on a smooth surface, base of trigger housing down and to the left. Assemble hammer spring housing, hammer spring, and hammer spring plunger into one unit. Place plunger in its seat against hammer, *making sure that open side of hammer spring housing is toward safety*, and hold assembled parts in a raised position with left thumb and forefinger. With right hand insert trigger into trigger slot so that notch at curved rear surface of finger piece bears against rear of slot in trigger housing. Arrange parts together so that bottom shoulder of hammer



FIGURE 11.

spring housing rests in notch just below and forward of trigger pin hole and the two wings of hammer spring housing straddle sear pin. (See fig. 11.) Push down with left thumb on top of rear end of hammer spring housing. (See fig. 12.) At the same time pull trigger rearward with right forefinger and push forward on sear with right thumb, thus compressing hammer spring and bringing trigger pin hole into alinement with holes in trigger housing. Hold this alinement with right hand bracing base of trigger housing against a firm surface. With left hand insert trigger pin

up to its head. Seat head of trigger pin fully while oscillating trigger assembly with right hand.

(3) *Clip latch.*—Place clip latch with clip latch spring attached in position on left side of receiver and start clip latch pin in from the front. Press rear end of clip latch to relieve tension of clip latch spring and push clip latch pin fully home. (Protrusion of pin head will result in a damaged stock.)

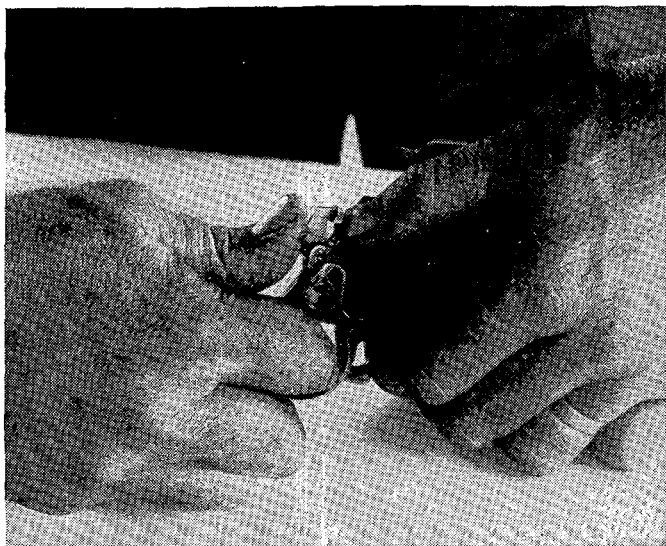


FIGURE 12.

(4) *Rear sight.*—Insert front lip of rear sight cover in slot at forward end of recess in receiver. Spring rear lip into its slot by pressing forward on rear surface. Insert forward end of rear sight base in opening in rear sight cover. Press it forward into position and slide rear sight base to the rear. Insert aperture in slot opening in rear sight base. Slide aperture to its extreme forward position. Holding rear sight base forward against rear sight cover, insert elevating pinion through left side of receiver, taking care that it meshes with

segment on aperture. Insert windage knob through right side of receiver and screw into rear sight base until zero mark on rear sight base registers with center line of windage scale on receiver. Assemble rear sight nut lock spring and nut lock and place them in position in windage knob around threaded end of elevating pinion, flange of nut lock outside, *taking care that flat cut on elevating pinion is alined with flat on nut lock.* Using combination tool, screw rear sight nut onto elevating pinion until tension desired on both elevating knob and windage knob is obtained. (If nut is screwed too tight, knobs become locked and cannot be turned.) Lower aperture as far as it will go by turning elevating pinion. Replace elevating knob and rotate it to position noted before disassembling (approximately 100 yards). Holding it in this position replace elevating knob screws and tighten.

(5) *Bolt assembly.*—Insert firing pin into bolt, making sure that tang enters slot in rear of bolt. Grasp bolt in left hand, top up, extractor recess to right, holding firing pin in place with little finger. Insert ejector with ejector spring attached. Insert extractor spring and plunger. Set stud of extractor into its hole in the bolt without forcing it against ejector spring. Place drift of combination tool in left groove of bolt with ejector in cut on face of tool. Press down on combination tool so as to compress ejector spring and aline ejector, then with thumb of left hand push extractor in until extractor plunger is seated. This operation should be performed only with the combination tool.

(6) *Barrel and receiver group.*—(a) Incline barrel and receiver assembly at an angle of approximately 45° , sights up and muzzle up and to the front. Hold bolt by right locking lug so that front end of bolt is slightly above and to the right of its extreme forward position in receiver. Insert rear end in its bearing on bridge of receiver rotating it in a counterclockwise direction sufficiently to permit tang of firing pin to clear top of bridge. Then guide left locking lug of bolt into its groove at a point just to rear of lug on left side of receiver, and right locking lug onto its bearing in receiver, and slide bolt back to its extreme rear position.

(b) Turn barrel and receiver assembly in left hand until barrel is down. With right hand grasp operating rod at

handle. Hold handle up and insert piston head into gas cylinder about $\frac{3}{8}$ of an inch, making sure that operating rod handle is to the left of receiver. Hold barrel and receiver assembly in left hand and rotate it to the right until barrel is uppermost. With right hand, adjust operating rod so that camming recess on its rear end fits over operating lug of bolt. Press operator rod forward and downward until the bolt is seated in its forward position.

(c) Place barrel and receiver assembly on a smooth surface, barrel down and muzzle to the left. Replace follower (with follower slide attached) so that its guide ribs fit into their grooves in receiver, square hole in follower to the right. Follower slide will rest on bottom surface of bolt when follower is in position correctly.

(d) With left hand replace bullet guide so that shoulders of bullet guide fit in their slots in receiver and hole in projecting lug is in line with holes in receiver.

(e) With left hand replace follower arm by passing its studded end through slot in bullet guide and inserting studs in their grooves in front end of follower. Place forked end of follower arm in position astride projecting lug on bullet guide with the pin holes in alinement. Insert rear arm of operating rod catch through clearance cut in bullet guide, making sure that its rear end is underneath forward stud on clip latch which projects into receiver opening. Aline holes in operating rod catch, follower arm, and bullet guide with those in receiver. Insert follower arm pin in side of receiver which is toward body and press pin home.

(f) If separated, insert operating rod spring into operating rod. Assemble follower rod and compensating spring by grasping spring in left hand and inserting follower rod with right hand, twisting two together so that compensating spring is fully seated on follower rod. Grasp knurled portion of the follower rod with thumb and forefinger of left hand, forked end to right, hump down. Insert left end of follower rod into operating rod spring, push to left and seat forked end against studs on follower arm. Hump on follower rod must be in slot in operating rod catch.

(7) *Three main groups.*—Insert U-shaped flange of stock ferrule in its seat in lower band. Pivoting about this point,

guide and press barrel and receiver group into position in stock. Insert trigger housing group, with trigger guard in its open position, into opening in stock. Press it into position and close and latch trigger guard.

b. Test the assembly as follows: Pull back and hold operating handle to its rearmost position; press down on follower and allow bolt to go fully home; set safety in its rearmost position; pull trigger; hammer should *not* fall; set safety in its foremost position; pull trigger; hammer *should* fall.

SECTION III

CARE AND CLEANING

■ 10. IN GARRISON AND CAMP.—a. Care and cleaning in garrison and camp include the care of the rifle necessary to preserve its condition and appearance during the periods when no firing is being done. Rifles in the hands of troops should be inspected daily to insure proper condition and cleanliness. Training schedules should allow proper time for cleaning rifles on each day when rifles are used in training.

b. The bore of the rifle will always be cleaned with a cleaning rod from the muzzle. The cleaning rod specified in SNL B-21 is of sufficient length to permit cleaning of the bore without damage to the follower or face of bolt. *If a longer rod is used the follower and face of bolt must be protected.* Numerous methods have been devised and found satisfactory. The simplest method is to block the follower and cover face of bolt with a piece of cloth stuffed into the receiver. To clean the bore, assemble a cloth patch to the cleaning rod. Insert the rod into the bore at the muzzle and move it forward and back several times, and remove the patch. **CAUTION:** In cleaning the bore, care must be taken not to foul the cleaning patch in the gas port. Repeat until several successive patches come out absolutely clean. Saturate a patch in sperm oil and push it through the bore, holding the rifle, top up, so that some sperm oil will flow into the gas port.

NOTE.—Sperm oil should be used when available. When not available, motor oil, weight 20, or any light grade machine oil may be used in an emergency. Aircraft lubrication oil is not a good preservative.

c. To clean the screw heads and crevices, use a small cleaning brush or small stick. To clean the metal surfaces, wipe with a dry cloth to remove moisture, perspiration, and dirt, then wipe with a cloth containing a small quantity of sperm oil. This protective film will be maintained *at all times*. To clean the outer surfaces of the rifle, wipe off the dirt with a lightly oiled cloth and clean with a soft dry one.

d. After cleaning and protecting the rifle as described above, place it in the gun rack without covering and without a plug in the muzzle or bore. Muzzle covers, gun covers, rack covers, and plugs must not be used because they cause sweating and promote rust. However, when squad rooms are being swept, the gun racks may be covered to protect the rifles from dust. Covers must be removed after the rooms have been swept.

■ 11. PREPARATORY TO FIRING.—a. The care and cleaning of the rifle preparatory to firing differs from the procedure prescribed in paragraph 10 in that lubricating grease (U. S. A. Spec. SXS77) is substituted for oil on many of the moving parts of the weapon. The grease is now issued in a collapsible tube. To apply the grease, rub a small quantity on a corner of a cleaning patch and apply it to the parts. Avoid excess quantities.

b. The following procedure will be observed to assure efficient functioning of the rifle:

- (1) Dismount main groups.
- (2) Clean and oil the bore very lightly. *Do not oil the chamber.*
- (3) Remove any carbon on the gas cylinder plug and piston head.
- (4) Thoroughly clean and lightly oil all metal parts. In cold weather use aircraft lubrication oil. In hot weather use sperm oil.
- (5) Apply a thin, uniform coating of lubricating grease to the parts listed below:
 - (a) Bolt lugs (locking and operating).
 - (b) Bolt guides.
 - (c) Cocking cam on bolt.
 - (d) Compensating spring.

- (e) Contact surfaces of barrel and operating rod.
- (f) Operating rod cam.
- (g) Operating rod guide groove in receiver.
- (h) Operating rod spring.

CAUTION: Do not apply lubricating grease to the follower slide or the under surface of the bolt, as the introduction of grease into the chamber may lead to the generation of excessive pressure.

(6) Assemble rifle and rub all outer surfaces with a lightly oiled rag to remove dust.

■ 12. AFTER FIRING.—The bores of all rifles must be thoroughly cleaned by the evening of the day on which they are fired. They should be cleaned in the same manner for the next 3 days. **CAUTION:** Under no circumstances will metal fouling solution be used in this rifle.

a. Immediate cleaning after firing, or as soon as possible.—Hold the rifle, bottom side up, so that no water will enter the gas port. Run several wet patches through bore. For this purpose water must be used; warm water is good, but warm, soapy water is better. Remove the patch section from the M3 cleaning rod and substitute the cleaning brush therefor; run the cleaning rod with brush assembled back and forth through the bore several times. Care should be used to insure that the brush goes all the way through the bore before the direction is reversed. Remove the brush and run several wet patches through the bore, removing them from the breech end. Follow this by dry patches until the patches come out clean and dry. Saturate a patch in sperm oil and push it through the bore, holding the rifle, top side up, so that some of the oil will flow into the gas port. **Caution:** In cleaning the bore, care must be taken not to foul the cleaning patch in the gas port.

b. Complete cleaning.—This cleaning should be done as soon as possible after that described in *a* above. If the rifle is to be fired the next day proceed as in paragraph 11. If the rifle is not to be fired in the next few days repeat procedure in *a* above for 3 days. In addition, the following instructions will be observed:

(1) *Chamber.*—Insert the cleaning patch deep into the slot of the chamber cleaning tool. Lay the tool with patch into

the palm of the left hand, close the left hand over the tool, and give the tool about three turns to the right. This will cause the patch to be neatly wrapped around the tool and will cover all its metal surfaces. A slight pressure with the forefinger of the left hand, while turning the tool, will twist the end of the patch much like the finished end of a hand-rolled cigarette, causing the patch to be sufficiently secured to the tool to permit neat entry into the chamber. This insures cleaning the full length of the chamber, prevents the patch from being crushed down to the lower end of the tool, and eliminates the danger of causing rings in the chamber by exposed portions of the cleaning tool. Clean by twisting the tool with patch in the chamber.

(2) *Gas cylinder and gas cylinder plug.*—Carbon will accumulate due to firing. The frequency of carbon removal is a factor peculiar to individual rifles. Excess deposits of carbon in the rifle manifest themselves by sluggishness in action and failure to feed.

(a) *Spline type.*—To remove accumulated deposits of carbon from the gas cylinder, remove the lock screw and remove carbon, using the screwdriver blade of the combination tool. The gas cylinder lock may be removed and the lock screw reinserted in the gas cylinder and threaded in enough to break loose the carbon. The inside of the gas cylinder should be thoroughly wiped clean and oiled at the conclusion of firing. (A few drops of oil placed between the rear gas cylinder lug and the operating rod, with the muzzle tipped down, will be sufficient if firing is contemplated on the next day. Hand operate the rod through a few cycles to distribute the oil properly.) The exterior finish should be cleaned and lightly oiled. The sight should be kept free of dust and dirt.

(b) *Screw-on type.*—Scrape the carbon from the exposed surface of the front of the gas cylinder and gas cylinder plug and piston head after extensive firing. Clean the gas cylinder plug and the grooves in the gas cylinder to insure correct seating of the plug. The frequency of this cleaning depends on the amount of firing. A sharp blade instrument, similar to a mess kit knife, should be used to remove the carbon from the gas cylinder plug and piston head. If an abrasive cloth is used care should be taken that the corners of the plug or

piston head are not rounded. *Do not remove the gas cylinder for cleaning.* The gas cylinder is cleaned by using the cleaning rod and a patch in the same manner that the bore is cleaned.

(3) *Exterior surfaces.*—Wipe off the exterior of the rifle with a dry cloth to remove dampness, dirt, and perspiration. Wipe all metal surfaces with sperm oil, the stock and handguards with raw linseed oil, and the sling with neat's-foot oil.

■ 13. ON THE RANGE OR IN THE FIELD.—The rifle must be kept clean and free from dirt and properly lubricated with lubricating grease. To obtain its maximum efficiency the following points must be observed:

a. Never fire a rifle with any dust, dirt, mud, or snow in the bore.

b. Keep the chamber clean and free from oil and dirt.

c. Never leave a patch, plug, or other obstruction in the chamber or bore. Neglect of this precaution may result in serious injury.

d. If the rifle gives indications of lack of lubrication and excessive friction, apply additional lubricating grease to the parts listed in paragraph 11 a (5). Excessive friction exists if the empty cases are being ejected to the right rear, and grease should be applied at the first opportunity as failures to feed and eject will occur if the condition is not corrected.

e. Keep a light coating of oil on all other metal parts.

f. Remove the carbon from the gas cylinder plug and the piston head when necessary.

g. In general it should not be necessary to remove any of the parts of the rifle in the field for cleaning except the trigger housing group and the gas cylinder plug. However, if the mechanism becomes very dirty the rifle may be disassembled into its three main groups and the necessary cleaning and lubricating accomplished.

h. During range firing, a selected and qualified man should be placed in charge of the cleaning of rifles at the cleaning racks or tables.

■ 14. PREPARATION FOR STORAGE.—a. Sperm oil is the most suitable oil for preserving the mechanism of rifles. This oil

is efficient for preserving the polished surfaces, the bore, and the chamber for a period of from 2 to 6 weeks, dependent on the climatic and storage conditions.

b. Rust preventive compound, medium, is a semisolid material. This compound is efficient for preserving the polished surfaces, the bore, and the chamber for a period of 1 year or less, dependent on the climatic and storage condition.

c. The rifles should be cleaned and prepared with particular care. The bore, all parts of the mechanism, and the exterior of the rifles should be thoroughly cleaned and then dried completely with rags. In damp climates, particular care must be taken to see that the rags are dry. After drying a metal part, the bare hands should not touch that part. All metal parts should then be coated either with sperm oil or rust-preventive compound, depending on the length of storage. (See *a* and *b* above). Application of the rust-preventive compound to the bore of the rifle is best done by dipping the cleaning brush in rust-preventive compound and running it through the bore two or three times. Before placing the rifle in the packing chest see that the bolt is in its forward position and that the firing pin is released. Then, handling the rifle by the stock and hand guard only, it should be placed in the packing chest, the wooden supports at the butt and muzzle having previously been painted with rust-preventive compound. Under no circumstances should a rifle be placed in storage contained in a cloth or other cover or with a plug in the bore. Such articles collect moisture which causes the weapon to rust.

■ 15. CLEANING OF RIFLES AS RECEIVED FROM STORAGE.—*a*. Rifles which have been stored in accordance with paragraph 14 will be coated with either sperm oil or medium rust preventive compound. Rifles received from ordnance storage will, in general, be coated with heavy rust preventive compound. Use a light oil or dry cleaning solvent to remove all traces of the compound or oil, particular care being taken that all recesses in which springs or plungers operate are cleaned thoroughly. After using the dry cleaning solvent make sure it is completely removed from all parts. Then follow instructions contained in paragraph 10. If the rifle

is to be fired immediately, follow instructions contained in paragraph 11.

NOTE.—Failure to clean the firing pin and the recess in the bolt in which it operates may result in gun failure at normal temperatures, and will most certainly result in serious malfunctions if the rifles are operated in low temperature areas, as rust preventive compound and other foreign matter will cause the lubricating oil to congeal or frost on the mechanism.

b. Dry cleaning solvent is a petroleum distillate, of low inflammability and noncorrosive, used for removing grease. It is generally applied with rag swabs to large parts and as a bath for small parts. The surfaces must be thoroughly dried immediately after removal of the solvent. To avoid leaving finger marks, which are ordinarily acid and induce corrosion, gloves should be worn by persons handling parts after such cleaning. Dry cleaning solvent will attack and discolor rubber.

SECTION IV

FUNCTIONING

■ 16. OBJECT.—This section is designed to provide a non-technical description of the functioning of the rifle. The object of instruction in this subject should be to lead the soldier to an understanding of the simple functioning of his weapon without emphasis on memorizing the matter of the text.

■ 17. WHEN TAKEN UP.—Instruction in functioning will be taken up after instruction in the disassembly, assembly, care and cleaning of the rifle.

■ 18. USE OF DUMMY CARTRIDGES.—The corrugated type of dummy cartridge (cal. .30, M1906) may be used for instruction in functioning. The use of the slotted type of dummy cartridge (range, cal. .30, M1) is prohibited. Special care must be exercised in the use of dummy cartridges that they do not introduce dirt or grit into the chamber of the rifle.

■ 19. DESCRIPTION OF CYCLE.—a. *Rearward movement.*—(1) When the rifle is loaded and the bolt closed, the hammer spring is compressed and the trigger lugs are engaged in the hammer hooks, holding the hammer in the cocked position. If pressure is then applied to the trigger, the trigger lugs are

disengaged from the hammer hooks, and the released hammer, actuated by the hammer spring, rotates about the hammer pin and strikes the firing pin which transmits the blow to the primer of the cartridge. (See fig. 13.) The bolt, however,

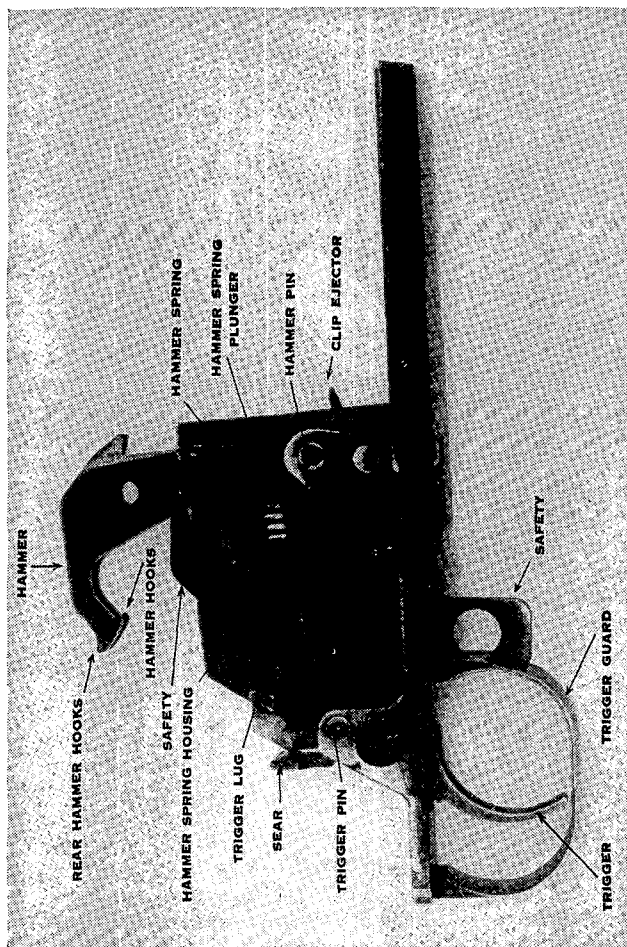


FIGURE 13.—Trigger housing group assembly with hammer in fired position.

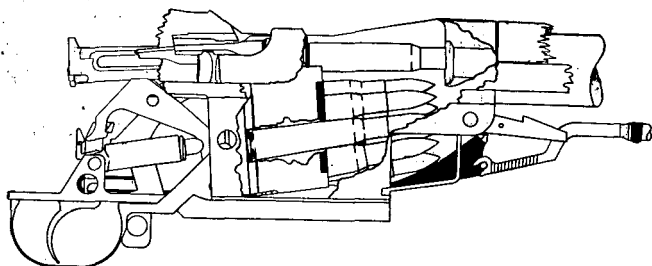
must be fully rotated to its locked position before this action can take place as the tang of the firing pin is blocked by the bridge of the receiver except when this tang is alined with the slot in the bridge. Moreover, until the bolt is rotated to its locked position, the hammer is prevented from striking the firing pin by the bolt camming lug striking the cam surface cut into the rear face of the bolt. (The bolt camming lug projects from the face of the hammer.) The shape of this cam surface is such that the bolt will be fully closed by a positive cam action when the hammer is released. In addition the safety must be in its foremost position so that it does not block the hammer and trigger.

(2) When the bullet passes the gas port some of the gas passes through it into the cylinder, where it strikes the piston end of the operating rod with sufficient force to drive the operating rod to the rear and compress the operating rod and compensating springs.

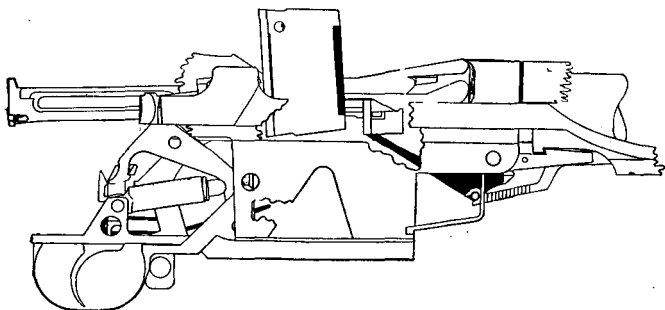
(3) The initial movement of the operating rod to the rear for about $\frac{5}{16}$ inch, is independent of the bolt mechanism, the operating lug merely sliding in the straight section of the recess in the operating rod. The cam surface of this recess then comes in contact with the operating lug and cams it up, rotating the bolt counterclockwise and disengaging the locking lugs on the bolt from the corresponding recesses in the receiver. This delay between the initial movement of the operating rod and the unlocking of the bolt permits the bullet to leave the muzzle, thus relieving the pressure in the barrel before the bolt is opened. The rotation of the bolt also cams the hammer back from the firing pin and withdraws the firing pin point into the bolt.

(4) As the operating rod continues its movement to the rear it carries the bolt which slides along the receiver. The empty cartridge case is withdrawn from the chamber by the extractor. (See fig. 14.) When the mouth of the empty cartridge case clears the breech, the ejector, which is continually pressing on the base of the cartridge, ejects the empty case to the right front through the action of the compressed ejector spring. The rear end of the bolt forces the hammer back and rides over it, thus compressing the hammer spring, and finally comes to rest near the rear end of the receiver.

With the bolt at its extreme rearward position the magazine is uncovered. The follower, actuated by the follower arm and the follower rod which transmit pressure from the operating rod spring and the compensating spring, then forces the cartridges upward in the clip so that the top cartridge



**BOLT MOVING TO REAR —
PIECE HAS BEEN FIRED.**



**BOLT IS IN REARMOST POSITION —
LAST ROUND IN CLIP WAS FIRED
AND EJECTED, CLIP IS BEING EJECTED.**

FIGURE 14.—Rearward movement.

lies in the path of the bolt. The rearward movement of the operating rod terminates when the rear end of its broad curved section contacts the front face of the receiver.

b. Forward movement.—(1) As the bolt moves forward, actuated by the compressed operating rod and compensating springs, the lower front face of the bolt comes in contact

with the base of the top cartridge and slides it forward into the chamber. The hammer, under pressure from the hammer spring, rides on the bottom of the bolt and tends to follow it, but is caught and held by the trigger lugs which engage the hammer hooks, if the pressure on the trigger has been released. (See fig. 15.) If, however, the pressure on the trigger has *not* been released, that is, if the trigger is held back after firing, the sear will engage the *rear* hammer hooks. (See fig. 16.) Subsequent release of the trigger disengages the sear from the hammer which then slides into engagement with the trigger lugs.

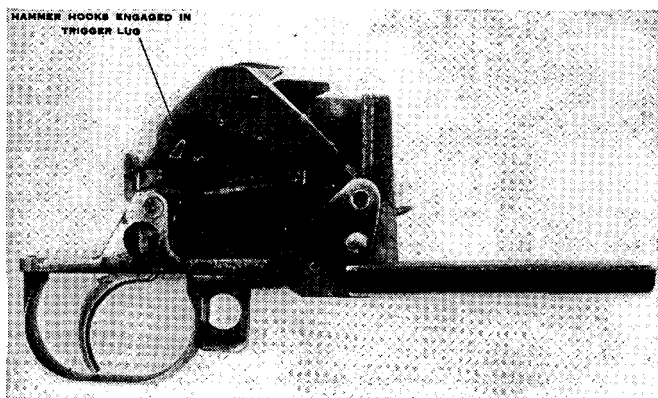


FIGURE 15.—Trigger housing group assembly with hammer in cocked position.

(2) When the bolt approaches its forward position, the rim of the cartridge is engaged by the extractor and the base of the cartridge forces the ejector into the bolt thus compressing the ejector spring. The operating lug is cammed downward by the rear surface of the cam recess in the operating rod, and in this manner the operating rod rotates the bolt clockwise to engage the locking lugs in the receiver. This section locks the bolt. The operating rod then continues to move forward for about $\frac{5}{16}$ inch until the rear end of the straight section of the recess in the operating rod contacts the operating lug on the bolt. The com-

compensating spring prevents the hooks on the follower rod from being disengaged from the follower arm during the forward movement. The rifle is then ready to be fired again. The cycle described above is repeated each time the trigger is squeezed except for the last round in the clip.

c. *Automatic ejection of empty cartridge clip.*—When the last round of a clip is fired, the operations of unlocking and opening the bolt and extracting and ejecting the empty case are accomplished as already described. The bolt clears the top of the receiver in its movement to the rear, and since

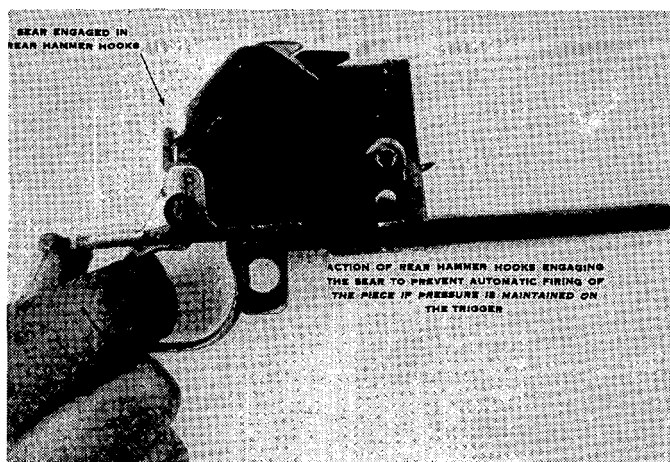


FIGURE 16.—Trigger housing group assembly with hammer hooks engaging the sear to prevent automatic firing.

the clip is empty the follower is pushed up to its extreme top position by the action of the follower arm, follower rod, compensating spring, and operating rod spring. The position of the follower rod under these conditions is such that it cams the forward end of the operating rod catch upward to engage the notch in the operating rod. At the same time the rear arm of the operating rod catch pivots about the follower arm pin and forces down the stud on the front end of the clip latch, thus rotating the clip latch and disen-

gaging it from the clip. Thereupon the empty clip is ejected by the clip ejector. (See fig. 14.) With the operating rod held to the rear by the operating rod catch, and the receiver empty, a loaded clip may be inserted without delay and firing resumed.

d. Automatic release of operating rod catch.—The automatic release of the operating rod catch, which allows the bolt to close when a loaded clip is inserted in the receiver, is accomplished in the following manner: When the loaded clip is forced down in the receiver it depresses the follower. As the follower nears the bottom of the receiver, the front end of the follower arm contacts and rotates the accelerator. The accelerator, in rotating about its pin in the operating rod catch, bears on a lug on the bullet guide and forces down the front end of the operating rod catch, thus releasing the operating rod. The operating rod is then moved forward by the pressure of the operating rod and compensating springs. At the same time the rear arm of the operating rod catch swings upward and releases the stud on the front end of the clip latch. The clip latch then rotates under the pressure of its spring, the rear lug of the clip latch moving inward to engage the notch in the clip and retain the clip in the magazine.

SECTION V

OPERATION

■ 20. OBJECT.—This section is designed to give the soldier that instruction necessary for the operation of the rifle.

■ 21. WHEN TAKEN UP.—The operation of the rifle will be taken up at any convenient time after instruction in care and cleaning has been completed.

■ 22. USE OF DUMMY CARTRIDGES.—As prescribed in paragraph 18.

■ 23. TO LOAD CARTRIDGE CLIP.—Insert eight rounds in the cartridge clip so that the base of each cartridge is close to the rear wall of the clip and the inner rib of the clip engages the extractor groove in the cartridge. It is immaterial whether the uppermost cartridge of the loaded clip is on the

left or right side as the follower slide adjusts itself for either loading. Experience shows, however, that for ease in inserting the clip the soldier prefers to have the uppermost cartridge on the right side of the clip.

■ 24. To LOAD RIFLE.—*a.* The operation of loading is performed with the piece locked, i. e., with the safety of the piece in its rearmost position except in sustained firing. Hold the rifle at the balance in the left hand. With the forefinger of the right hand, pull the operating rod handle smartly to the rear until the operating rod is caught by the operating rod catch. With the right hand take a fully loaded clip and place it on top of the follower. Place the right side of the right hand against the operating rod handle and with the thumb of the right hand press the clip down into the receiver until it engages the clip latch. Swing the thumb to the right so as to clear the bolt in its forward movement and release the operating rod handle. The closing of the bolt may be assisted by a push forward on the operating rod handle with the heel of the right hand. The technique of loading the rifle properly is readily acquired after performing the operation a few times with dummy cartridges in accordance with these instructions.

b. The loading and reloading of the rifle without hurried movements and consequent fumbling will be demonstrated and taught to all men under instruction.

■ 25. To UNLOAD RIFLE.—*a.* To unload a cartridge from the chamber, hook the right thumb over the operating rod handle, pull and hold the operating rod in the extreme rear position, thus extracting and ejecting the round.

b. To remove the loaded clip from the receiver, hold the rifle with the right hand, thumb on operating rod handle, fingers around the trigger guard. Place the fingers of the left hand over the receiver and press in on the clip latch with the left thumb. The clip will then be ejected upward from the receiver and into the left hand. (See fig. 17.) **CAUTION:** Do not allow the bolt to move forward during the operation as it will push the top cartridge forward and prevent normal ejection of the clip.

c. To close the bolt on an empty chamber and retain the loaded clip in the receiver, press down on the top cartridge

in the clip and allow the bolt to slide forward, making sure that it is fully closed.

■ 26. TO OPERATE RIFLE AS SINGLE LOADER.—The receiver being empty, pull the operating rod to the rear until it is caught by the operating rod catch. With the right hand, place one round in the chamber, seating it in place with the thumb. With the right side of the right hand against the operating rod handle and the fingers extended and joined,



FIGURE 17.—To unload.

force the operating rod handle slightly to the rear, depress the follower with the right thumb, and permit the bolt to ride forward about 1 inch over the follower. Then remove the thumb from the follower, release the operating rod handle and push forward on the operating rod handle with the heel of the hand to be certain that the bolt is completely closed. No type of ammunition will be loaded *into the receiver* except in full clips.

■ 27. **TO FIRE RIFLE.**—Squeeze the trigger for each shot. When the eighth shot has been fired, the empty clip is automatically ejected and the bolt remains open.

■ 28. **TO SET RIFLE AT SAFE.**—The loaded rifle will be kept locked until the moment for firing. To make this adjustment set the safety in its rearmost position inside the trigger guard. In this position the trigger cannot be pulled as the upper end of the trigger is blocked from disengaging from the hammer hooks, and the hammer is held in the cocked position by the hook on the safety being engaged with the lug on the left side of the hammer. The rifle may be loaded and operated by hand when *locked* but cannot be fired. To *unlock* the rifle set the safety in its foremost position.

■ 29. **TO ADJUST REAR SIGHT.**—The rear sight is adjusted for range by turning the elevating knob on the left side. This knob has numbered graduations for 200, 400, 600, 800, 1,000, and 1,200 yards of range and index lines between these graduations for 100, 300, 500, 700, 900, and 1,100 yards. Adjustment for windage is made by turning the windage knob on the right. Each windage graduation represents an angular adjustment of 4 minutes. Both elevating and windage knobs are provided with “clicks” which represent approximately 1 minute of angle or 1 inch on the target for each 100 yards of range. Arrows on the knobs indicate the direction in which to turn them to secure corresponding changes in the point of impact of the bullet. Rotation of the elevating knob may be eased by forcing the knob outward (away from receiver) while turning.

■ 30. **SAFETY PRECAUTIONS.**—The soldier must be impressed with the fact that while any cartridges remain in the receiver, after a round has been fired, the rifle is ready to fire. The gun is safe only when it is “cleared”; in other words, the gun is never *known* to be *safe* when the bolt is closed.

■ 31. **TO CLEAR RIFLE.**—*a.* To clear rifle, pull the operating rod fully to the rear, extracting and ejecting the cartridge from the chamber. Remove the clip from the magazine as described in paragraph 25 and *leave the bolt open.*

b. In range firing, whenever firing ceases, execute clear rifle as prescribed above.

SECTION VI

IMMEDIATE ACTION AND STOPPAGES

■ 32. OBJECT.—This section is designed to provide necessary instruction in the related subjects of immediate action and stoppages.

■ 33. WHEN TAKEN UP.—Instruction in immediate action and stoppages will be completed before any firing is done by the individual.

■ 34. IMMEDIATE ACTION.—*a. General.*—Immediate action is the unhesitating application of a probable remedy for a stoppage. Immediate action deals with the method of reducing stoppages and not the cause. It is taught as an unhesitating manual operation to be applied to reduce stoppages without detailed consideration of their causes.

b. Procedure.—(1) If the loaded rifle fails to fire when the trigger is pulled, count to "20" to allow for a hang fire, and then pull the operating handle to its rearmost position ejecting the round. Release the operating handle and if the bolt goes fully home aim and fire.

(2) If the bolt cannot be fully locked in *a* above, pull the operating rod handle to the rear. Check for a battered round, dirt, or obstruction on the face of the bolt, in the chamber, or in the locking lug recess. Discard the battered round; remove the obstruction. Reload, aim, and fire.

(3) The rifle fires but fails to feed. Keep the rifle in action by working the operating rod handle as it is still an effective combat weapon. A detailed examination for the malfunction may be made later when time permits.

(4) The above procedure of immediate action will almost invariably keep the rifle in action. In case this immediate action is not successful, a more detailed examination for the possible malfunctions listed in paragraph 35*e* may be made as circumstances permit.

■ 35. STOPPAGES.—*a. General.*—While immediate action and stoppages are closely related as to subject matter, the former is treated separately to emphasize its importance as an automatic and definite procedure to be applied to overcome stoppages. Proper care of the rifle before, during, and after firing will usually eliminate stoppages. Stoppages which

cannot be remedied by the application of immediate action can best be eliminated if the soldier has an understanding of the functioning of the weapon and the causes of stoppages.

b. Failure to fire.—(1) *Causes.*—Failures to fire are generally caused by—

(a) Defective ammunition.

(b) Defective firing pin.

(c) Bolt not fully closed when hammer strikes firing pin.

(2) *Action.*—(a) If the primer of a round is deeply indented, the round is defective. Discard the round. If the primer is not indented or but very lightly indented, the firing pin may be short or broken or the bolt may not have been fully closed. Check for dirt or some obstruction which does not permit the bolt to lock fully. Remove the obstruction. If the rifle is clean and lubricated, check the firing pin. Replace it if defective.

(b) *Removal of a broken firing pin.*—If the piece fails to fire and the operating handle cannot be moved to the rear by a sharp blow with the heel of the hand, the firing pin may be broken, and having come out of its seat in the bolt it may have become wedged between the rear of the bolt and the top of the receiver. Remove the trigger housing. Generally the firing pin will fall out. If it does not fall out, separate the barrel and receiver group from the stock group and remove the firing pin. If the trigger housing cannot be removed easily, *do not force it out of its recess*. The firing pin is caught under the lug on the left of the hammer. Open the trigger guard as far as it will open without force. Turn the rifle, barrel down, and shake it, at the same time oscillating the trigger guard until the trigger housing can be removed. Remove the firing pin as indicated above.

c. Failure to feed.—(1) *Types.*—Failures to feed are caused by failure of the bolt to go far enough to the rear to pick up a new round. A failure to feed may have any one of a number of causes. It will generally result in one of the following types of stoppages:

(a) Those in which the bolt fails to go fully home.

(b) Those in which the bolt does go fully home.

(2) *Action to reduce stoppage of the first type.*—Stoppages of the first type may be caused by a battered round, dirt in the

locking recesses, an obstruction on the face of the bolt, a dirty chamber, or a ruptured cartridge case part of which remains in the chamber. Remove the battered round, dirt, or other obstruction; clean the chamber; or remove the ruptured cartridge case as the case may be. Occasionally this stoppage may be caused by a clip which has lost its spring tension and does not hold the cartridge firmly in line. When this occurs, the cartridge will be found "cocked" in the gun with the nose of the bullet one side or the other of the entrance to the chamber. Remove the round; remove the clip and discard it.

(3) *Action to reduce stoppage of the second type.*—Occasionally, when a stoppage of the second type occurs, the spent case is not ejected but is re-fed back into the chamber. This condition is caused by lack of lubrication, excessive friction of the moving parts, or lack of sufficient gas pressure due to the formation of carbon in the gas port. In any case the bolt has not moved far enough to the rear to permit proper functioning. The conditions are remedied by removing all carbon and thoroughly lubricating all parts as prescribed in paragraphs 10 to 15, inclusive.

d. Failure to extract.—(1) *Causes.*—Failures to extract are generally caused by—

- (a) Extremely dirty chamber.
- (b) Extremely dirty ammunition.
- (c) Improper assembly of the rifle, i. e., failure to replace the extractor plunger and spring.
- (d) Cartridge case chambered in a hot barrel.
- (e) Broken extractor.

(2) *Action.*—(a) When a failure to extract occurs, the bolt may be found fully locked with a spent case in the chamber. Generally, most failures to extract can be remedied by pushing the operating rod fully forward and then pulling it smartly to the rear. If this does not remove the case, use the combination tool or cleaning rod.

(b) Sometimes the empty case will be left in the chamber, the extractor ripping through the base of the cartridge. When this occurs the bolt generally will attempt to feed a fresh cartridge into the chamber. It will then be necessary to remove this round before the spent case can be removed.

(c) Where a dirty chamber or dirty ammunition is indicated, clean the chamber and discard or clean very dirty

ammunition. Faulty assembly or a broken extractor will cause recurring failures to extract. Replace missing or broken parts.

e. Table of stoppages.—The following table may be found of value. It includes the stoppages outlined above and others that have not been covered but which may occur infrequently.

TABLE OF STOPPAGES

Malfunction	Cause	Correction by the soldier
Clip jumps out on seventh round.	Bent follower rod.....	Replace.
Failure to extract.....	(1) Dirty or rough chamber. (2) Restricted gas port....	(1) Clean chamber. (2) Clean gas port.
Failure to feed.....	(1) Dirty or rough chamber. (2) Restricted gas port.... (3) Dirty rifle or improperly lubricated. (4) Bent clip..... (5) Ruptured cartridge case in chamber.	(1) Clean chamber. (2) Clean gas port. (3) Clean rifle and lubricate. (4) Replace clip. (5) Remove ruptured cartridge case.
Fires automatically.....	Sear broken or remains in open position.	Replace trigger assembly or hammer spring housing.
Safety releases when pressure is applied on trigger.	Round heel on safety, or broken safety.	Replace safety.
Pull on trigger does not release hammer.	(1) Deformed hammer or trigger or worn trigger pin. (2) Trigger strikes trigger housing.	(1) Replace defective part. (2) Turn in to ordnance.
Hammer releases but gun does not fire.	(1) Bolt not all way seated. (2) Defective ammunition. (3) Broken firing pin.....	(1) Clean and lubricate. (2) Discard round. (3) Replace.
Rear sight elevation jumps.	Loose rear sight nut.....	Tighten.
Creep in trigger.....	Burs on trigger or hammer lugs.	Turn in to ordnance.

f. Other stoppages.—In the event of stoppages that are not mentioned above and that cannot be reduced, the rifle should be turned in for examination and repair.

SECTION VII

SPARE PARTS AND ACCESSORIES

■ 36. SPARE PARTS.—*a.* The parts of any rifle will in time become unserviceable through breakage or wear resulting from continuous usage, and for this reason spare parts are supplied. These are extra parts provided with the rifle for replacement of the parts most likely to fail; for use in making minor repairs; and in general care of the rifle. They should be kept clean and lightly oiled to prevent rust. Sets of spare parts should be kept complete at all times. Whenever a spare part is taken to replace a defective part in the rifle, the defective part should be repaired or a new one substituted in the spare part set as soon as possible. Parts that are carried complete should at all times be correctly assembled and ready for immediate insertion in the rifle. The allowance of spare parts is prescribed for the rifle in SNL B-21.

b. With the exception of replacements with the spare parts mentioned in *a* above, repairs or alterations to the rifle by using organizations are prohibited.

■ 37. APPENDAGES.—Appendages are items not required for use in the operation of the major equipment but are used attached to or in connection with such equipment. For the rifle they consist of the bayonet, M1905, and bayonet scabbard, M1910.

a. Bayonet.—The bayonet is a blade sharpened along the entire lower edge and partly along the top edge. The bayonet guard is constructed so as to fasten the bayonet securely to the rifle or its scabbard. Wooden assemblies on both sides of the tang provide a grip for use as a hand weapon.

b. Bayonet scabbard.—The scabbard, shaped to receive the bayonet, consists of a fabric-covered body with a leather reinforce at its tip which contains a drain hole. The scabbard is held to the belt of the soldier by two hooks.

■ 38. ACCESSORIES.—*a. General.*—Accessories include the tools required for assembling and disassembling and for the cleaning of the rifle; also the gun sling, spare parts containers, covers, arm locker, etc. Accessories should not be used for purposes other than those for which they are intended, and when not in use they should be stored in the places or receptacles provided for them. There are a number of accessories, the names or general characteristics of which indicate their uses or application. Therefore, detailed descriptions or methods of the uses of such items are not outlined herein. However, accessories of a special nature or those which have special uses are described in *b* to *h* below.

b. Arm locker and rack.—The arm locker and the arm rack are used to store or stack rifles and pistols to prevent mishandling or pilfering.

c. Barrel reflector.—This is a small, box-shaped device having a short tube which slips into the chamber of the rifle barrel. It also has a mirror and an opening through which the reflection of the bore is obtained. The condition of the rifle bore may thereby be readily determined.

d. Brush and thong.—The brush and thong are used for cleaning the bore of the rifle. The case is partitioned so that one end contains the oil and oil dropper and the other holds the tip, weight, thong, and brush.

e. Cleaning rod, M3, and cleaning brush, M2.—The cleaning rod, M3, has a handle at one end and is threaded at the other end to receive the patch section or the brush. This rod is of the correct length to prevent damaging the follower or the face of the bolt. The cleaning brush, M2, is used to clean the bore of the rifle.

f. Combination tool.—This tool consists of three parts—the chamber-cleaning tool, the handle, and the screw-driver blade. The slot in the chamber-cleaning tool is for attachment of a cleaning patch. The movable screw-driver blade is used for the gas cylinder plug screw and various other screws, while the notched blade of the handle is used on the rear sight nut. The small cylindrical projection is used to drift out pins. It is also used in conjunction with the V-shaped groove cut into the face of the handle to assemble the extractor and ejector. The curved undercut lug or

hook, commonly called the hand extractor, is used to extract a cartridge case after firing if the extractor should fail to extract the case.

g. Gun sling.—The gun sling, fastened to the loops of the rifle, is adjusted to suit the particular soldier using it. The sling consists of a long and short strap, either of which may be lengthened or shortened as desired.

h. Ruptured cartridge extractor.—The ruptured cartridge extractor has the general form of a caliber .30 cartridge. It consists of three parts—the spindle, the head, and the sleeve. To use the ruptured cartridge extractor, the cartridge clip and live cartridges must be removed from the rifle. The ruptured cartridge extractor is then inserted through the ruptured opening of the case and pushed forward into the chamber. The bolt is let forward without excessive shock so that the extractor of the rifle engages the head of the ruptured cartridge extractor. As the operating rod is drawn back, the ruptured cartridge extractor holding the cartridge on its sleeve is extracted.

SECTION VIII

INSTRUMENTS

■ 39. FIELD GLASS, TYPE EE.—The field glass, type EE (fig. 18), complete, consists of the field glass with its carrying case.

a. Description.—(1) The field glass consists of two compact prismatic telescopes (5) pivoted about a common hinge (4) which permits adjustment for interpupillary distances. A scale (3), graduated every 2 millimeters from 56 to 74, permits the observer rapidly to set the telescope to suit his eye distance when the spacing of his eyes is known. The eyepiece (1) can be focused independently for each eye by screwing in or out. Each is provided with a diopter scale (2) for rapid setting when the observer knows the correction for his eye. The zero graduations indicate the settings for normal eyes.

(2) The left telescope is fitted with a glass reticle (fig. 19 ① and ②) upon which are etched a vertical mil scale, a horizontal mil scale, and on field glasses of older manufacture

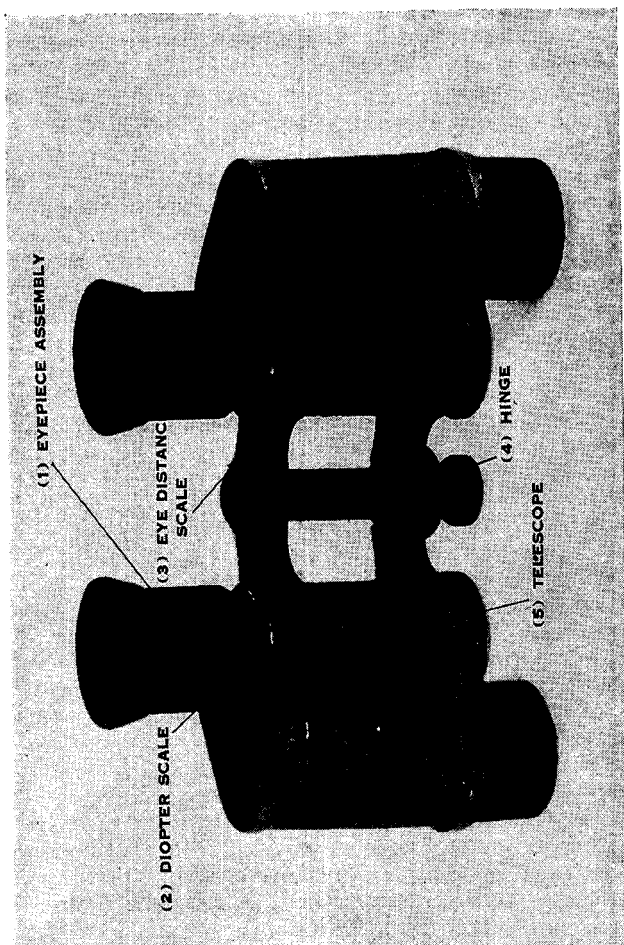


FIGURE 18.—Field glass, type EE.

a stadia graduated similarly to the sight leaf graduation on the service rifle, but inverted.

b. Use.—The field glass is used for observations and the measurement of small horizontal and vertical angles in mils. The vertical stadia scale (inverted sight leaf) is used to pick up auxiliary aiming marks in direct laying and to determine troop safety for overhead fire.

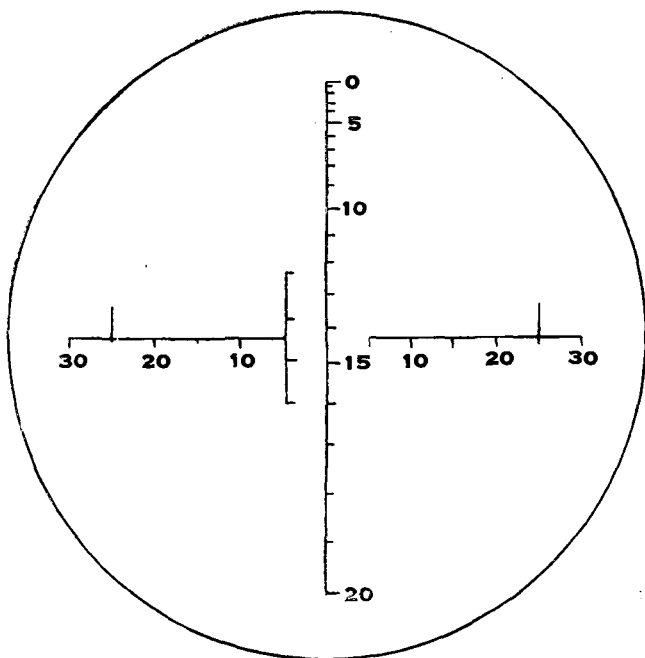
c. Preliminary adjustments; interpupillary distances.—To adjust the glass so that the eyepieces are the same distance apart as the pupils of the observer's eyes, point the glass at the sky and open or close the hinged joint until the field of view ceases to be two overlapping circles and appears to be one sharply defined circle. Note the reading on the scale (3), which indicates the spacing of the observer's eyes. The similar setting of any other field glass will then accommodate his eyes.

d. Focus of the eyepiece.—Look through the glasses, both eyes open, at an object several hundred yards away. Place the hand over the front of one telescope and screw the eyepiece of the other in or out until the object is sharply defined. Repeat this operation for the other eye and note the reading on the diopter scale. The similar setting of any other field glass will accommodate the eyes.

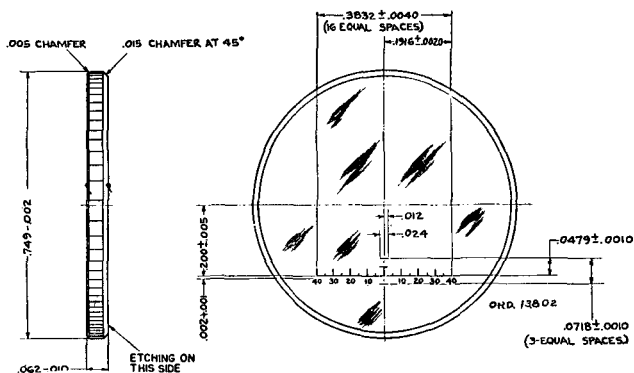
e. Operation.—(1) In using the glass it should be held in both hands and pressed lightly to the eyes so as to keep the relation with the eyes constant without transmitting tremors from the body. The bent thumbs should fit into the outer edges of the eye sockets in such a manner as to prevent light from entering in rear of the eyepieces. When possible it is best to use a rest for the glass or elbows.

(2) The mil scales are seen when looking through the glass, and by superimposing them upon the required objects the horizontal and vertical angles may be read between these objects.

f. Care.—The field glass is a rugged, serviceable instrument but should not be abused or roughly handled.



① Older manufacture.



② Recent manufacture.

FIGURE 19.—Reticles on field glasses.

SECTION IX

AMMUNITION

■ 40. **GENERAL.**—The information in this section pertaining to the several types of cartridges authorized for use in the U. S. rifle, caliber .30, M1, includes description, means of identification, care, use, and ballistic data.

■ 41. **CLASSIFICATION.**—*a.* (1) Based upon use, the principal classifications of ammunition for this rifle are—

(a) *Ball*—for use against personnel and light matériel targets.

(b) *Tracer*—for observation of fire and incendiary purposes.

(c) *Armor-piercing*—for use against armored vehicles, concrete shelters, and similar bullet-resisting targets.

(2) The following types are similar to ball, except for the following:

(a) *Armor-piercing* is painted black for $\frac{1}{4}$ inch from the point.

(b) *Tracer* is painted red for $\frac{1}{4}$ inch from point.

b. (1) Other types provided for special purposes are—

(a) *Guard*—for guard purposes (gallery practice cartridges also used for this purpose).

(b) *Blank*—for simulated fire, signaling, and salute.

(c) *Dummy*—for training.

(2) For guard and blank ammunition the chamber pressure is insufficient to operate self-loading mechanism. Single shots may be fired by pulling back the operating rod handle to extract the fired cartridge case.

■ 42. **LOT NUMBER.**—When ammunition is manufactured, an ammunition lot number is assigned which becomes an essential part of the marking in accordance with specifications. This lot number is marked on all packing containers and the identification card inclosed in each packing box. It is required for all purposes of record, including grading and use, reports on condition, functioning, and accidents, in which the ammunition might be involved. Those lots only of grades appropriate for the weapon will be fired. Since it is impracticable to mark the ammunition lot number on each individual cartridge, every effort should be made to

maintain the ammunition lot number with the cartridges once the cartridges are removed from their original packing. Cartridges which have been removed from the original packing and for which the ammunition lot number has been lost are placed in grade 3. It is therefore obvious that when cartridges are removed from their original packings they should be so marked that the ammunition lot number is preserved.

■ 43. GRADE.—Current grades of all existing lots of small arms ammunition are established by the Chief of Ordnance and are published in Ordnance Field Service Bulletin No. 3-5. No lot other than that appropriate to the weapon will be fired. Color bands painted on the sides and ends of the packing boxes further identify the various types of ammunition. The following color bands are used to identify cartridges:

Armor-piercing.....	Blue on yellow.
Ball.....	Red.
Blank.....	Blue.
Dummy.....	Green.
Gallery practice.....	Brown.
Guard.....	Orange.
Tracer.....	Green on yellow.

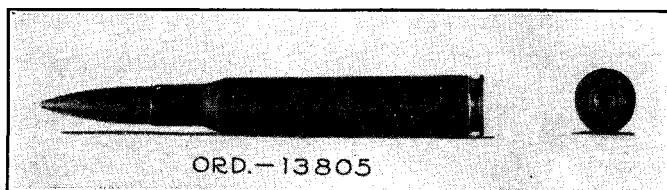
■ 44. CARE, HANDLING, AND PRESERVATION.—a. Small arms ammunition is not dangerous to handle. Care, however, must be observed to keep the boxes from becoming broken or damaged. All broken boxes must be immediately repaired, and careful attention should be given that all markings are transferred to the new parts of the box. The metal liner should be air tested and sealed if equipment for this work is available.

b. Ammunition boxes should not be opened until the ammunition is required for use. Ammunition removed from the airtight container, particularly in damp climates, is apt to corrode, thereby causing the ammunition to become unserviceable.

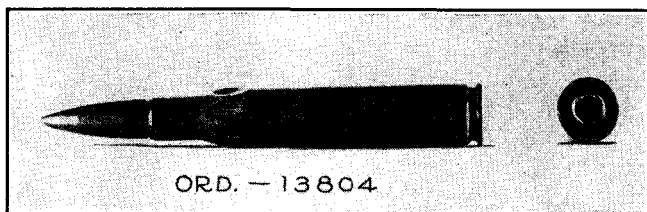
c. The ammunition should be protected from mud, sand, dirt, and water. If it gets wet or dirty, wipe it off at once. Light corrosion, if it forms on cartridges, should be wiped

off. However, cartridges should not be polished to make them look better or brighter.

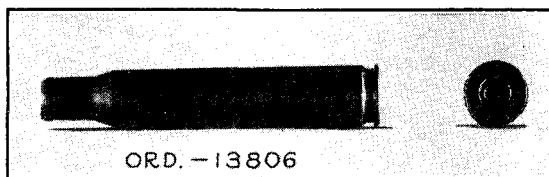
d. No caliber .30 ammunition, other than blanks, will be fired until it has been positively identified by ammunition lot number and grade.



① Ball, M2.



② Guard, M1906.



③ Blank, M1909.

FIGURE 20.—Cartridges, caliber .30.

■ 45. CARTRIDGE, BALL, CALIBER .30, M2 AND M1.—The approximate maximum range for the M2 cartridge is 3,450 yards, while for the M1 it is 5,500 yards. The M2 type is standard.

■ 46. PRECAUTIONS IN FIRING BLANK AMMUNITION.—*a.* It is dangerous to fire rifles loaded with blank cartridges at per-

sonnel representing an enemy at distances of less than 20 yards as the wad or paper cup may fail to break up.

b. Misfires in which the primer explodes but fails to ignite the powder charge may prove dangerous when blank ammunition is being fired. In this type of misfire, some of the powder may be blown into the bore of the weapon. A series of such rounds in which the powder fails to ignite due to moisture or other causes will result in an accumulation of powder sufficient to cause serious damage when ignited by a normal cartridge. When misfires are encountered in blank ammunition in excess of 5 percent, the firing of the lot will be suspended and reported to the Chief of Ordnance.